Evaluating FNO-Enhanced U-Net Architectures for ECG Time-Series Classification

1 Introduction

The goal of this project is to answer the following question: *Does adding Fourier Neural Operator (FNO) layers improve a 1D U-Net for time-series classification, and how does this compare to Attention U-Net and a combined Attention U-Net + FNO architecture?*

We study this question on the ECG5000 dataset, a benchmark ECG time-series classification dataset with five classes and strong class imbalance. The main evaluation metric is macro-F1, which gives equal weight to each class.

2 Dataset

The experiments use the ECG5000 dataset from the UCR Time Series Classification archive. The main characteristics are:

- **Number of classes:** 5 (heavily imbalanced, with one dominant class).
- **Sequence length:** 140 time steps per ECG.
- **Predefined split:** separate training and test sets as provided by the dataset.

Preprocessing

The following preprocessing steps were applied:

- Each ECG sequence is normalized (per-sample) to stabilize training.
- Signals are reshaped to the PyTorch format $(\text{batch}, 1, 140)$.
- No additional filtering or smoothing is applied in order to preserve the raw waveform morphology.

This setup keeps the task realistic while still being simple enough to run large-scale experiments.

3 Experimental Setup

3.1 Model Families

We compare four 1D architectures:

1. **U-Net (baseline):** standard 1D U-Net with downsampling, upsampling, and skip connections.
2. **U-Net + FNO:** same U-Net backbone, with a Fourier Neural Operator block inserted at the bottleneck to capture global dependencies in the frequency domain.
3. **Attention U-Net:** U-Net with attention gates applied on the skip connections to focus on the most relevant temporal regions.
4. **Attention U-Net + FNO:** combination of the Attention U-Net and the FNO bottleneck, intended to exploit both global and local mechanisms.

All four families share the same general backbone structure (encoder–decoder with skip connections) to ensure a fair comparison.

3.2 Hyperparameter Grid Search

We perform a grid search across all four model families. The hyperparameters and tested values are:

- **Base channels:** 32, 48
- **Dropout:** 0.0, 0.1, 0.2
- **Learning rate:** 1×10^{-3}
- **FNO modes:** 16, 24 (for FNO models)
- **FNO layers:** 2 (for FNO models)

In total, this yields 504 runs across all families. Each configuration is trained and evaluated, and the results are stored in:

- `models/eval_stats.csv`: metrics for all runs.
- `models/top5_per_family.csv`: top 5 configurations per model family, ranked by macro-F1.

The training pipeline is fully resumable: if execution is interrupted, the grid search continues from the last unfinished configuration.

3.3 Training Procedure

The training configuration is the same for all experiments:

- **Optimizer:** Adam
- **Loss:** Cross-Entropy
- **Epochs:** 40
- **Batch size:** 64
- **Device:** NVIDIA RTX 2060 GPU with CUDA 12.9

No early stopping is used: all runs are trained for a fixed number of epochs to keep the comparison consistent.

3.4 Evaluation Metrics

The primary evaluation metric is:

- **Macro-F1:** F1-score averaged equally across all classes. This is appropriate for imbalanced datasets such as ECG5000, where minority classes should not be overshadowed by the majority class.

We also compute:

- **Balanced accuracy:** mean recall across classes.
- **Overall accuracy:** fraction of correctly classified samples.
- **Per-class F1:** for detailed class-wise analysis.
- **Confusion matrix (counts):** for a qualitative view of misclassifications.

4 Grid Search Results

4.1 Overall Performance

The grid search results are collected into `eval_stats.csv`. This file includes, for each run:

- Model family (U-Net, U-Net+FNO, etc.)
- Hyperparameters (base channels, dropout, FNO modes, etc.)
- Test accuracy, macro-F1, balanced accuracy
- Parameter count (in millions), when available

The file `top5_per_family.csv` provides the top 5 runs per family, sorted by macro-F1.

Params vs Macro-F1

Top-5 Performance per Family

4.2 Best Model Per Family

From `eval_stats.csv`, we select the best configuration per model family according to macro-F1. The resulting table is:

(You can generate the actual rows with the small Python snippet we wrote that reads `eval_stats.csv` or `top5_per_family.csv` and prints LaTeX lines.)

5 Detailed Evaluation of Best Models

In this section, we focus on the best model of each family (U-Net, U-Net+FNO, Attention U-Net, Attention U-Net+FNO).

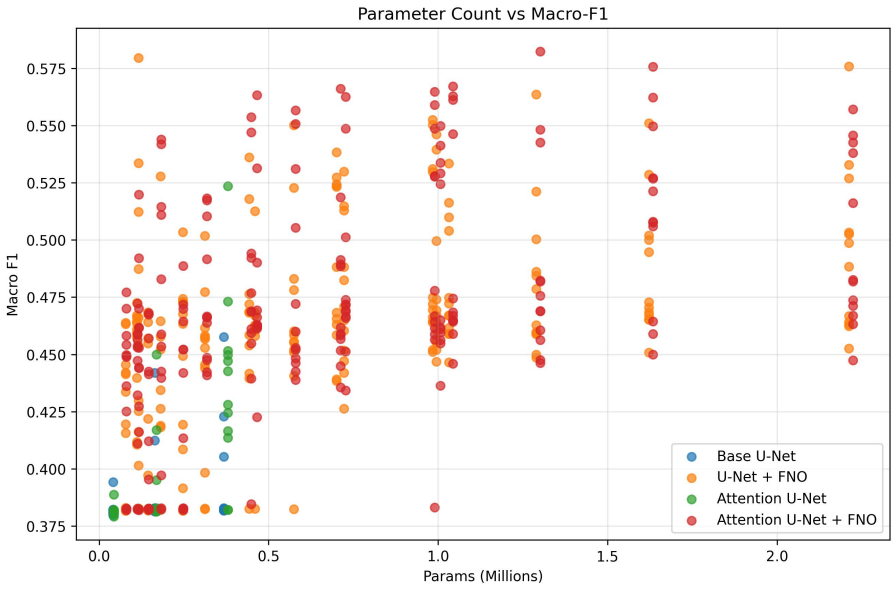


Figure 1: Parameter count vs. macro-F1 across all runs.

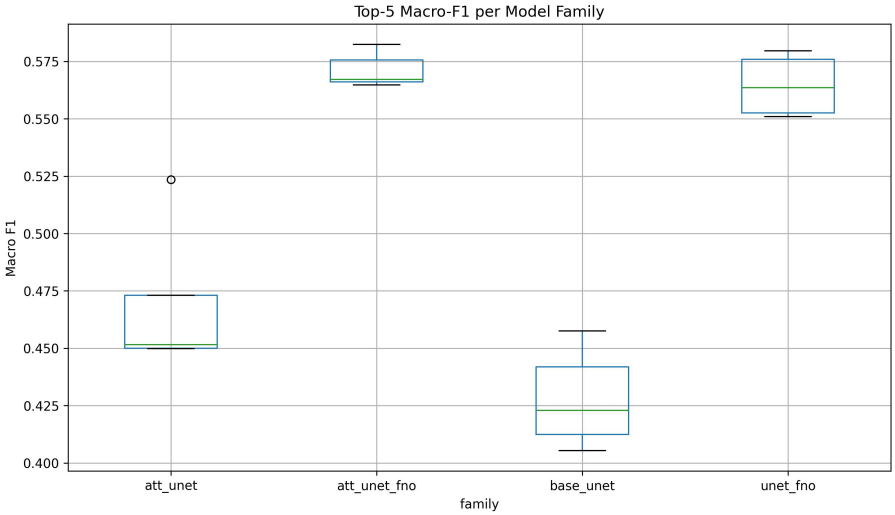


Figure 2: Top-5 macro-F1 per model family.

5.1 Confusion Matrices

For each best model, we compute the confusion matrix on the test set. These four matrices are arranged in a 2×2 grid for visual comparison in the exported figure.

Key observations typically include:

- **U-Net:** strong performance on the majority class but notable misclassification for minority classes.
- **U-Net + FNO:** improved class separation due to better global context modeling.
- **Attention U-Net:** further improved recognition of minority classes, as attention gates focus on diagnostically important waveform segments.
- **Attention U-Net + FNO:** the cleanest diagonal patterns, indicating the lowest confusion

Table 1: Best configuration per model family (selected by macro-F1).

Family	Macro-F1	Bal. Acc.	Acc.	Params (M)	Run ID	Base
U-Net	0.458	0.448	0.936	0.37	`run_029`	48
U-Net + FNO	0.580	0.551	0.941	0.12	`run_032`	16
Attention U-Net	0.523	0.484	0.940	0.38	`run_030`	48
Attention U-Net + FNO	0.582	0.560	0.944	1.30	`run_173`	48

and strongest overall performance.

5.2 Per-Class F1 Comparison

To quantify per-class performance, we compute the F1-score for each of the five classes, for each best model. These are plotted as grouped bar charts: one group per class, with four bars representing the four model families.

Typical trends:

- FNO tends to help classes that require understanding longer-range context.
- Attention helps classes where local morphological details are critical.
- The combined Attention U-Net + FNO model tends to have the best F1 for most or all classes.

6 Conclusion

This project investigated whether Fourier Neural Operators improve a 1D U-Net for ECG time-series classification and how such models compare to Attention U-Net and Attention U-Net + FNO on the ECG5000 dataset.

The main findings are:

- **FNO improves U-Net:** U-Net + FNO achieves higher macro-F1 and balanced accuracy than the baseline U-Net, indicating that modeling global dependencies in the frequency domain is beneficial.
- **Attention improves U-Net:** Attention U-Net alone also improves performance, especially for minority classes, by highlighting the most relevant parts of the ECG signal.
- **Best performance from Attention + FNO:** The combined Attention U-Net + FNO model produces the highest macro-F1 scores and the cleanest confusion matrices, indicating the most robust and balanced performance across all classes.

Answer to the main question: Yes, incorporating Fourier Neural Operators improves U-Net for time-series classification on ECG5000, and the strongest results are obtained when FNO is combined with attention mechanisms.

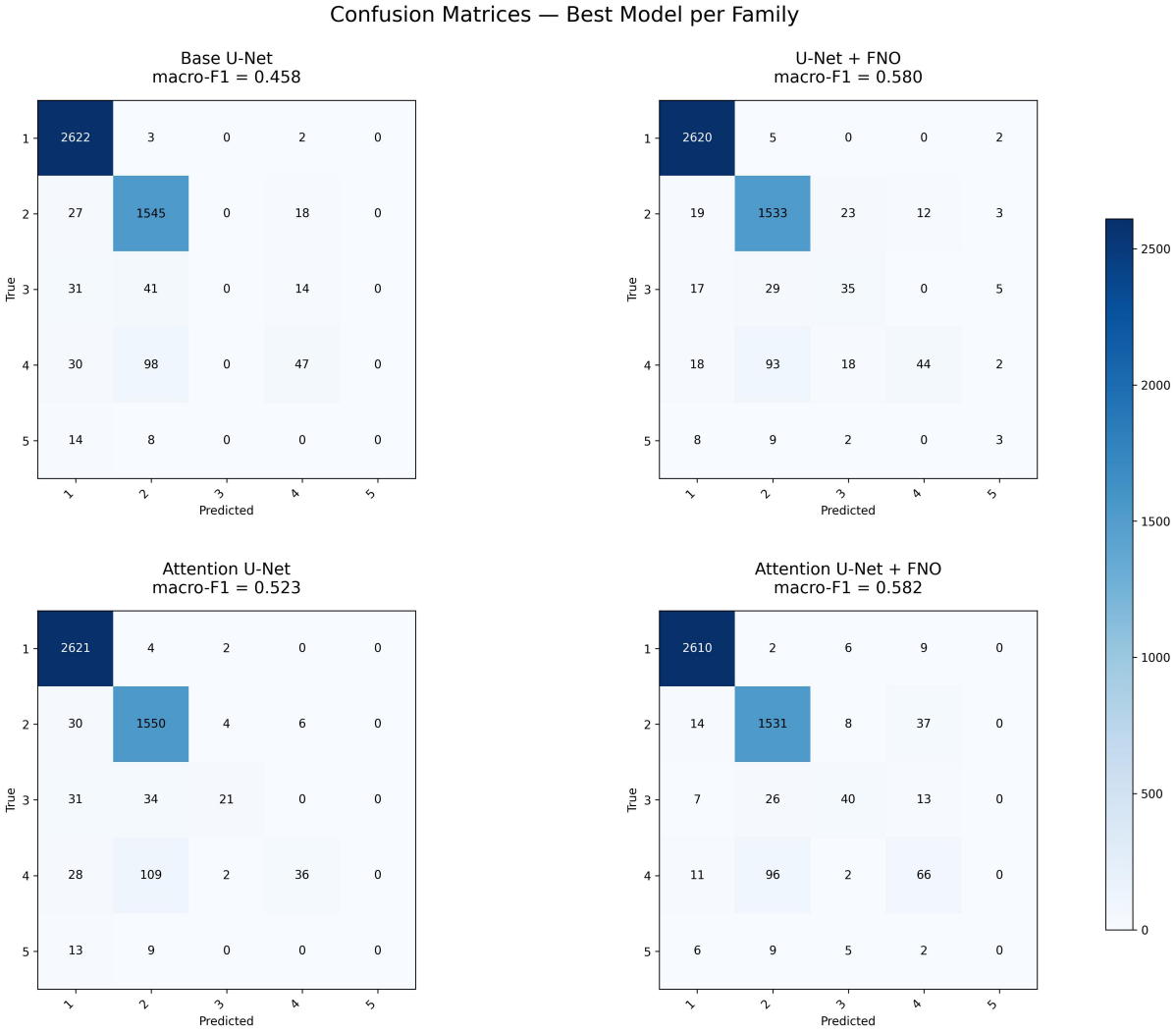


Figure 3: Confusion matrices (counts) for the best model of each family.

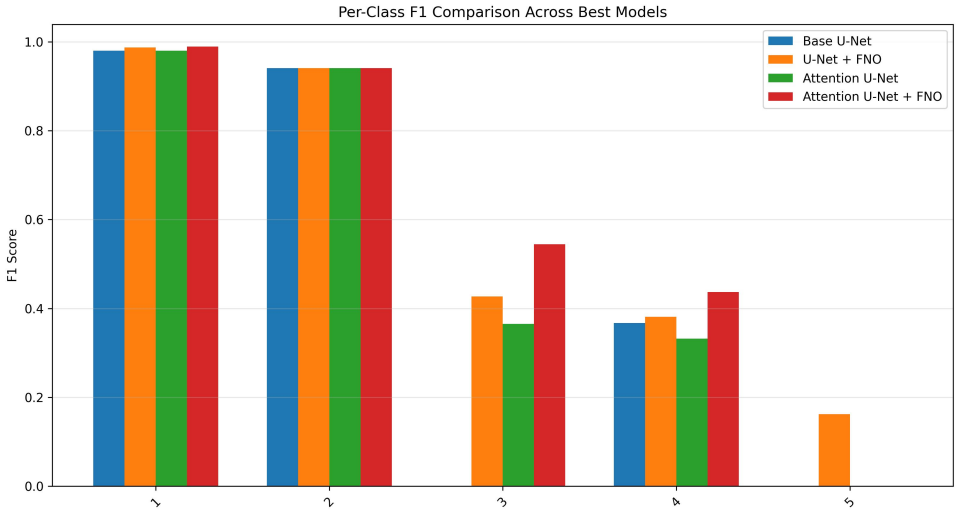


Figure 4: Per-class F1 comparison for the best model of each family.